

AEGISPAY — GROW

LangGraph growth agent · V3

How the AI safely grows a merchant's revenue. The AI decides what to propose; the control plane decides what may be authorized — clearly separated.

The flow

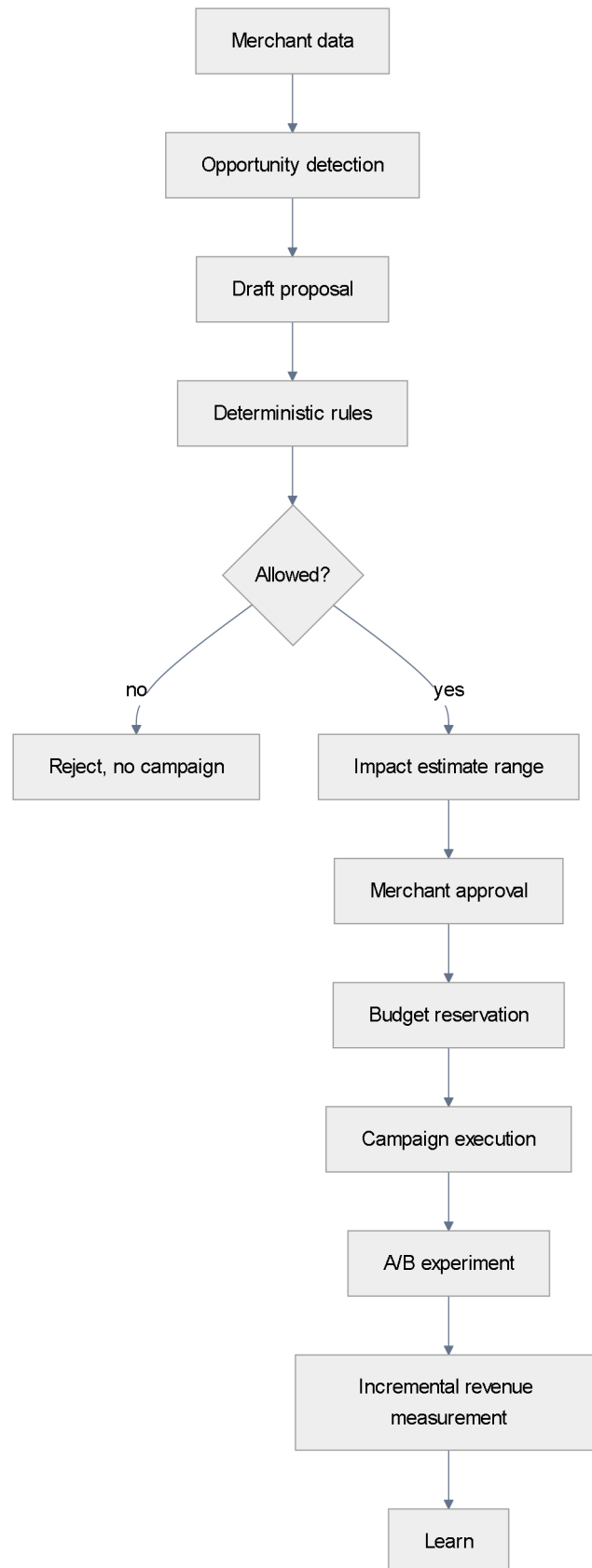


Figure 1 — From merchant data to a measured, budget-limited campaign

- 1 **Merchant data.** Grouped sales, no customer PII.
- 2 **Opportunity detection.** Real correlation (shoe → socks).
- 3 **Draft proposal.** The AI suggests an offer, budget, target.
- 4 **Deterministic rules.** Max discount, max budget, min margin, duration, frequency, allowed targeting.

- 5 **Impact estimate.** An honest range, labelled an estimate.
- 6 **Merchant approval.** The AI cannot start it alone.
- 7 **Budget reservation.** Atomic, so it can never overspend.
- 8 **Campaign execution.** Inside the reserved budget.
- 9 **A/B experiment.** Control vs treatment.
- 10 **Incremental revenue.** Test – control, not raw sales.
- 11 **Learn.** Repeat what works, drop what doesn't.

The rule that separates AI from money: the AI decides what to propose; the control plane decides what may be authorized.

Safety rails

Rail	Bound
Maximum discount	hard % cap
Maximum budget	hard ■ cap
Minimum margin	margin floor after discount
Campaign duration	days cap
Frequency cap	per-customer contact limit
Allowed targeting	value/behaviour only; no protected attributes
Atomic budget ledger	`spent + cost <= budget` in one operation
Kill switch	halt all campaign spend instantly

Budget ledger (atomic)

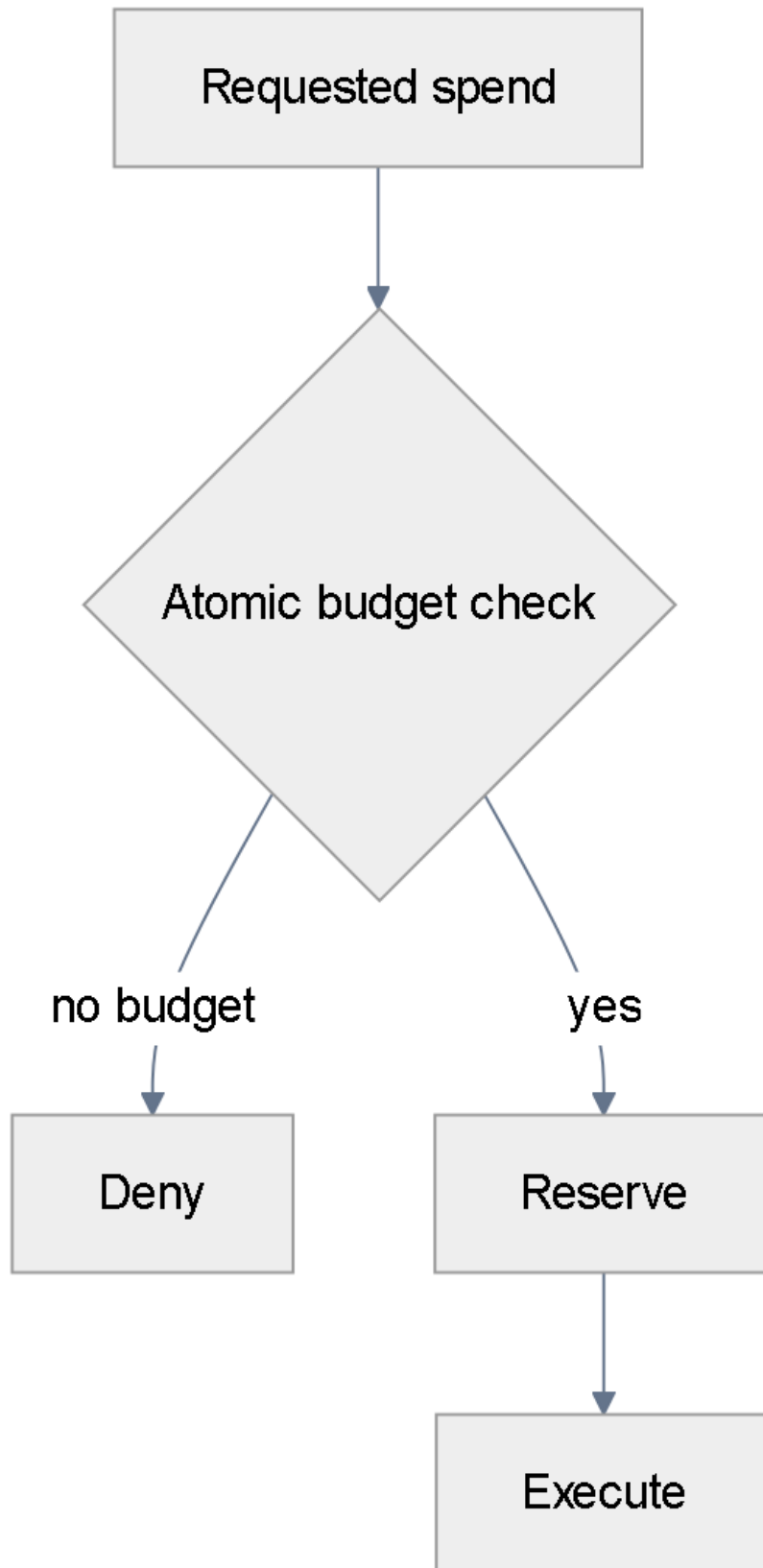


Figure 2 — Concurrent requests reserve budget atomically; no overspend and no double-spend

The AI cannot raise its own budget. Only a merchant/policy admin can, and that change is audited.

A/B + incremental measurement (GROW)

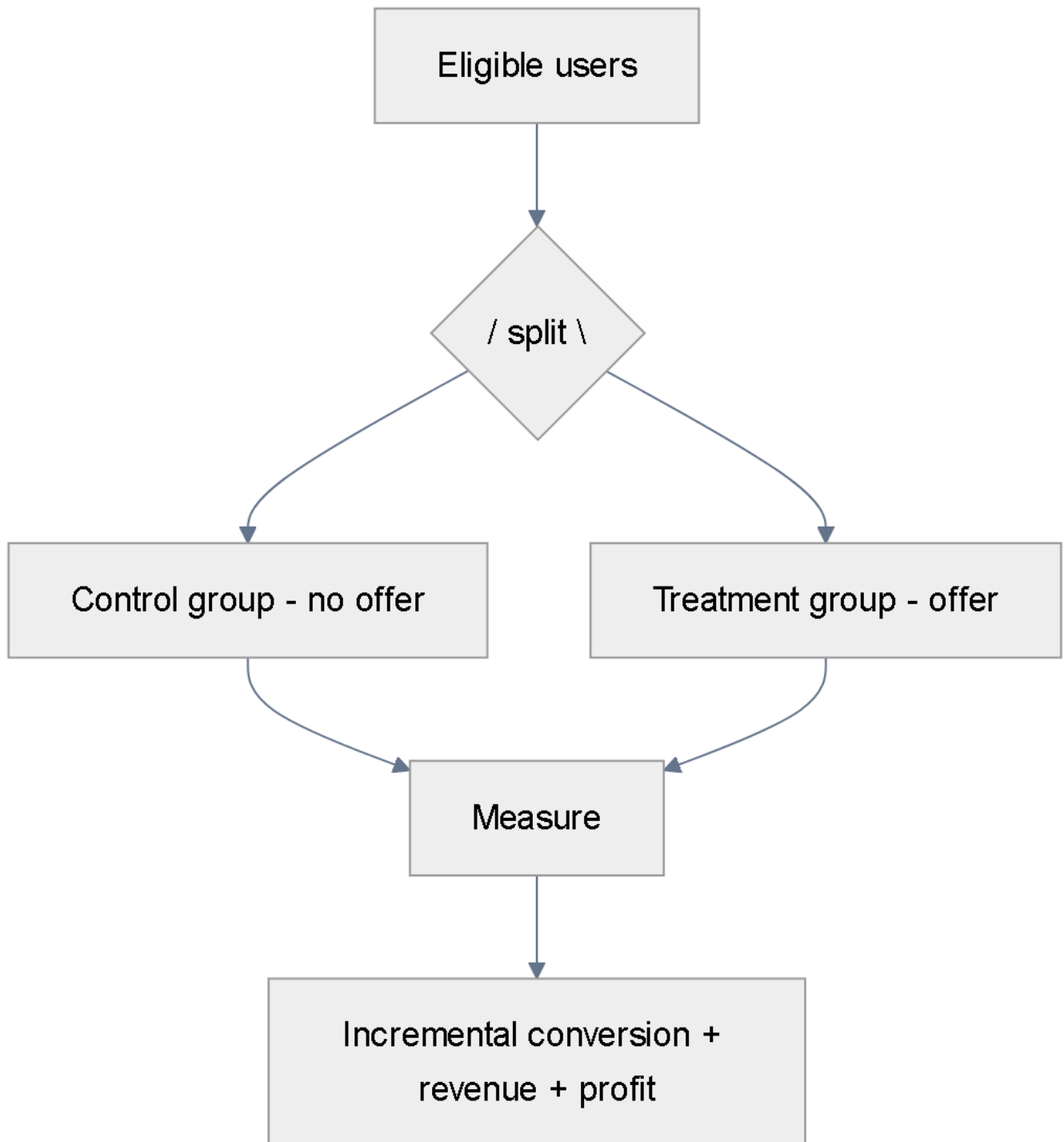


Figure 3 — Control vs treatment; measure the increment, not the campaign's headline

Campaign revenue is **not** automatically campaign-generated revenue — it must beat a control group.

Incremental uplift = treatment – control (conversion, revenue, profit).

Results are a range with a confidence level, labelled an estimate.

The AI is only scaled if the measured effect is real and positive.

One realistic failure

Step	Result
AI proposes a campaign budget	■20,000
Merchant policy allows	■5,000
Deterministic rules	reject the proposal
Campaign starts?	no
Audit record	campaign.rejected · reason: budget exceeds cap

The AI made a suggestion. Policy said no. No money moved, and the rejection is recorded. That is the point.

Why AegisPay is different

AI can reason and recommend, but only the deterministic AegisPay control plane validates, authorizes and controls financial actions. Razorpay executes only approved actions.

No direct money path. The AI runtime has no payment keys, no DB credentials, no unrestricted money tools.

Deterministic authorization. Policy, risk and a scoped, expiring authorization gate every action.

Provider truth. Payment finality comes only from a verified webhook or reconciliation — never a guess.

Prevents, not promises. Idempotency + UNKNOWN-never-blind-retry are designed to prevent duplicate financial effects.

Tamper-evident. Every decision is recorded in a hash-chained, anchored audit ledger and Transaction Passport.

Simple to run. Two services, PostgreSQL + Redis + SQS on ECS — no unnecessary complexity.